

SMART CROSS Frontend Documentation

1. Project Overview

Smart Cross is a railway crossing information application designed to help users know the current status of a railway crossing and estimate when it will close or open.

The application displays four dynamic crossing states:

Open

Closing Soon

Closed

Opening Soon

These states are displayed on the same main dashboard screen and change dynamically according to the current crossing status.

2. Objectives

The main objectives of the frontend are:

Provide clear and real-time crossing status information.

Show the estimated closing or opening time.

Display countdown information when required.

Allow users to select and view different railway crossings.

Provide a simple and user-friendly interface.

Provide notifications for important crossing status changes.

Provide a map to locate railway crossings.

Ensure the application is responsive on different screen sizes.

3. Frontend Screens

The Smart Cross frontend consists of the following main screens:

Landing Page

Crossing Selection

Main Dashboard / Crossing Status

Map

Notifications

Settings

The Open, Closing Soon, Closed, and Opening Soon conditions are not separate screens. They are different dynamic states of the Main Dashboard.

4. Landing Page

The Landing Page is the first screen users see when they open the application.

Main Elements

Smart Cross logo

Application name

Short tagline

Brief description of the application

Get Started button

Background/visual representing a railway crossing

Functionality

The Get Started button takes the user to the Crossing Selection screen.

5. Crossing Selection Screen

This screen allows users to select the railway crossing they want to check.

Main Elements

Search bar

List of available railway crossings

Crossing name

Crossing location

Distance from the user, if available

Select/View Status button

Functionality

Users can search for a crossing and select it to view its current status on the Main Dashboard.

6. Main Dashboard / Crossing Status

The Main Dashboard is the core screen of the Smart Cross application.

It displays the current status of the selected railway crossing. The interface remains the same while the status information changes dynamically.

Common Elements

Crossing name

Crossing location

Current status

Status indicator

Estimated time

Countdown timer

Next expected event

Last updated time

Refresh/Live status indicator

6.1 Open State

This state is displayed when the railway crossing is currently open.

Information Displayed

Status: OPEN

Crossing is currently available for users to cross.

Estimated time when the crossing may close.

Next expected event.

Example

OPEN

Crossing is currently open.

Expected to close at: 10:42 PM

6.2 Closing Soon State

This state is displayed when the railway crossing is expected to close soon.

Information Displayed

Status: CLOSING SOON

Estimated closing time

Countdown until closure

Warning message

Example

CLOSING SOON

The crossing is expected to close at 10:42 PM.

Closing in: 08 minutes

6.3 Closed State

This state is displayed when the railway crossing is currently closed.

Information Displayed

Status: CLOSED

Estimated opening time

Estimated waiting time

Next expected event

Example

CLOSED

The crossing is currently closed.

Expected to open at: 10:58 PM

Estimated waiting time: 16 minutes

6.4 Opening Soon State

This state is displayed when the railway crossing is expected to open soon.

Information Displayed

Status: OPENING SOON

Estimated opening time

Countdown until opening

Information about the upcoming opening

Example

OPENING SOON

The crossing is expected to open at 10:58 PM.

Opening in: 02 minutes

7. Map Screen

The Map screen helps users locate railway crossings and view their status geographically.

Main Elements

Interactive map

User's current location

Railway crossing markers

Crossing names

Crossing status indicators

Distance from the user

Functionality

Users can select a crossing directly from the map and view its current status.

8. Notifications Screen

The Notifications screen displays important updates related to railway crossings.

Notifications May Include

Crossing is closing soon

Crossing has been closed

Crossing is opening soon

Crossing is now open

Changes in estimated opening or closing time

Main Elements

Notification title

Notification message

Time/date

Read/unread status

9. Settings Screen

The Settings screen allows users to manage application preferences.

Main Elements

Notification settings

Location settings

Default crossing

Theme settings

About Smart Cross

App information

10. Loading and Error States

Loading and error states are handled within the existing screens and are not considered separate screens.

Loading State

Displayed while the application is fetching crossing information.

Example:

Fetching latest crossing status...

Error State

Displayed when crossing information cannot be retrieved.

Example:

Unable to fetch crossing status.

Try Again

11. Frontend Navigation

The basic navigation flow of the application is:

Landing Page → Crossing Selection → Main Dashboard

From the Main Dashboard, users can navigate to:

Map

Notifications

Settings

Navigation Structure

Landing Page ↓ Crossing Selection ↓ Main Dashboard ■ ↓ ■ Map | Notifications | Settings

12. Responsive Design

The Smart Cross frontend will be designed to work across different devices.

Supported Screen Sizes

Mobile phones

Tablets

Laptops

Desktop computers

The main status information should remain clearly visible on smaller screens.

Mobile Priority

On mobile devices, the most important information should be displayed in the following order:

Crossing Name

Current Status

Estimated Time

Countdown

Next Expected Event

Additional Information

13. Reusable Frontend Components

The frontend will use reusable components to maintain consistency and simplify development.

Main Components

Navbar

Status Card

Crossing Selector

Countdown Timer

Crossing Information Card

Map Component

Notification Card

Alert Banner

Loading Component

Error Component

Footer

14. Frontend and Backend Interaction

The frontend will receive railway crossing information from the backend/API.

The data may include:

Crossing name

Location

Current status

Estimated closing time

Estimated opening time

Next expected event

Last updated time

The frontend will use this information to determine which status should be displayed.

Status Flow

Backend Data → Status Processing → Main Dashboard → Dynamic Status Display

The dashboard will display only the current applicable state at a given time.

For example:

Open → Closing Soon → Closed → Opening Soon → Open

The interface will update as the crossing status changes.

15. Conclusion

The Smart Cross frontend is designed to provide users with a simple and clear way to understand the current and upcoming status of railway crossings.

The Main Dashboard acts as the central screen, dynamically displaying one of four states: Open, Closing Soon, Closed, or Opening Soon, along with the estimated time and relevant crossing information.

The overall frontend focuses on clarity, real-time information, easy navigation, responsiveness, and a user-friendly experience.